

Prototypes

Here we'll take a look at some products that are currently in development, and which, hopefully, should reach the markets, and you, within the next few years. Instead of looking at the available prototypes individually, we'll look at some of the possible implementations of paper displays, and see what products are being developed to fill those gaps.

Electronic Newspapers / Books

This is the very first implementation that the world thought up, which, unfortunately, is still only in the prototype stage. The most obvious implementation of a paper display is to replace paper! Since the most common interaction people have with actual paper is perhaps the newspaper, it is only natural that paper displays try and find their way into this space.

How can a paper display replace your newspaper? The answer is simple. Consider this: you probably go out and buy a newspaper everyday, or wait for the delivery boy to drop off one at your home. What happens if you have to leave home early on certain days? What about the news that happened after the paper was printed? What about those of us who like to read three and four different newspapers? Though TV and the Internet have fulfilled some of these needs, it's still not enough!

Now think of a paper display that's the same size as your newspaper. It's foldable, for easy storage, just like your newspaper, and isn't 40 pages thick! Now imagine if you could connect this paper via Wi-Fi or WiMax to news servers. Essentially, you would just need the single paper display to read all the news from all your favourite news sources. Obviously, news distribution would have to become subscription services, much like iTunes.

Not only do paper displays in this form solve the problem of storage and mobility, it also means that this sort of a system would enable you to access news as it happened, much like RSS feeds do for you right now. The added advantage here would be that you can sit and read it like a newspaper, which beats peering at a monitor any day!

Another advantage is that you could essentially sort your news according to taste. What

this means is that, say, you prefer *The Times Of India* for their sports news, *The Indian Express* for their political news and *The Hindustan Times* for general news, you could program your paper display to show you the relevant news from the desired news source. This is akin to using a news site that allows you to personalise the way your news appears, except that using the paper display you could be anywhere at all—travelling in a train, standing on a street corner waiting for a bus, or even while in the loo!

So what products have been, or are, being developed in this context?

Sony LIBRIé

Though only available in Japan right now, this is an e-book reader that uses a paper display. It can store up to 10 MB of e-books, and boasts of an impressively sharp resolution of 170 pixels per inch. The display, unlike all other displays available on readers or PDAs, is absolutely clearly readable even in direct sunlight. Though the screen has a dull plastic look, it's almost as readable as newsprint.



The Sony LIBRIé is not exactly the perfect e-reader, but it has a great contrast ratio for reading e-books, even under direct sunlight

The reader runs on four AAA batteries, which can last for as much as 10,000 page changes. Thankfully, the option of AAA batteries means that you can just pick up a fresh set at almost any general store across the world, and not have to rely on a power outlet to recharge any internal batteries!

There's a memory stick slot to add additional memory to the LIBRIé, and a small speaker and headphone outlet to play any embedded audio in your e-books. It also features a keyboard that lets you use the inbuilt dictionary, so that you can translate words into English. We're hoping to see a fully English version soon!

There are drawbacks though, namely screen refresh rates—or the lack thereof! When you click to turn a page, it takes a second to refresh the screen. While it may seem that we're being unreasonably harsh by complaining about this, try entering text on a display that takes a second to show every letter you type! Hopefully, these kinks will be ironed out soon—perhaps when a "LIBRIé 2" is launched!



This is a wall-sized paper display that was set up at an exhibition in Japan. It used E-Ink's technology to display newspaper articles on the paper display, while images were displayed on the smaller LCD